

6263-ITAI-1371-Intro to Machine Learning

Midterm Exam: Project Proposal and Dataset Description

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1. Dataset URL Source

The raw data is sourced directly from the official Kaggle repository:

URL: <https://kaggle.com>

(<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>)

2. Description of the Dataset

The Telco Customer Churn dataset contains 7,043 rows and 21 features capturing consumer behavior metrics from a telecommunications corporation. The target attribute is "Churn" (indicating whether a customer left the company within the last month: Yes or No). The data features consist of a critical mix of demographic domains (gender, senior citizen status, partner, dependents), subscribed services (phone service, multiple lines, internet, online security, streaming options), and financial account metrics (tenure duration, contract type, payment method, monthly charges, and total charges).

3. Machine Learning Problem and Objectives

This dataset presents a classic binary classification machine learning problem. For the upcoming Final Exam, the objective is to train and optimize predictive classification algorithms (such as Logistic Regression, Decision Trees, and Random Forests) to accurately identify customers at high risk of churning. From a business intelligence perspective, the operational goal is to maximize the model's "Recall" metric, ensuring the company flags as many potential churning accounts as possible to implement targeted retention campaigns and reduce revenue deficit.

4. Proposed Data Preprocessing Plan (Modules 4 & 5 Compliance)

To stabilize the predictive capabilities of the final machine learning frameworks, the training data partition (70% split) will undergo rigorous programmatic transformation within Python to address inherent dataset anomalies:

- **Handling Missing Entries:** The "TotalCharges" column contains structural empty string spaces. These will be programmatically imputed utilizing the feature median value to prevent bias.
- **Categorical Encoding:** Multiclass text fields, including "Contract" and "PaymentMethod," will be converted to algorithmic numbers using One-Hot Encoding to avoid non-ordinal ranking weight penalties.
- **Feature Scaling:** Disparate mathematical boundaries between features like "tenure" (months scale) and "MonthlyCharges" (dollar scale) will be standardized via StandardScaler to achieve a normal distribution.
- **Class Balancing:** The target labels suffer from a heavy class imbalance (mostly non-churning customers). We will programmatically resolve this by executing statistical minority upsampling to prevent the machine learning models from building a systematic prediction bias toward the majority class.